

PITCH DEPOWER CONTROL CAMPAIGN (V4)

High-Fidelity Dynamic Multi-Body Space-Frame Optimization

Windswept & Interesting Ltd — windswept.energy

Engineering Source-of-Truth Document

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Campaign Scope: 128 High-Resolution Simulations | Multi-Threaded Execution

Diagnostics: Column Buckling Location, Transverse Whipping Accel, Rotational Jerk, and Tether Location Mapping

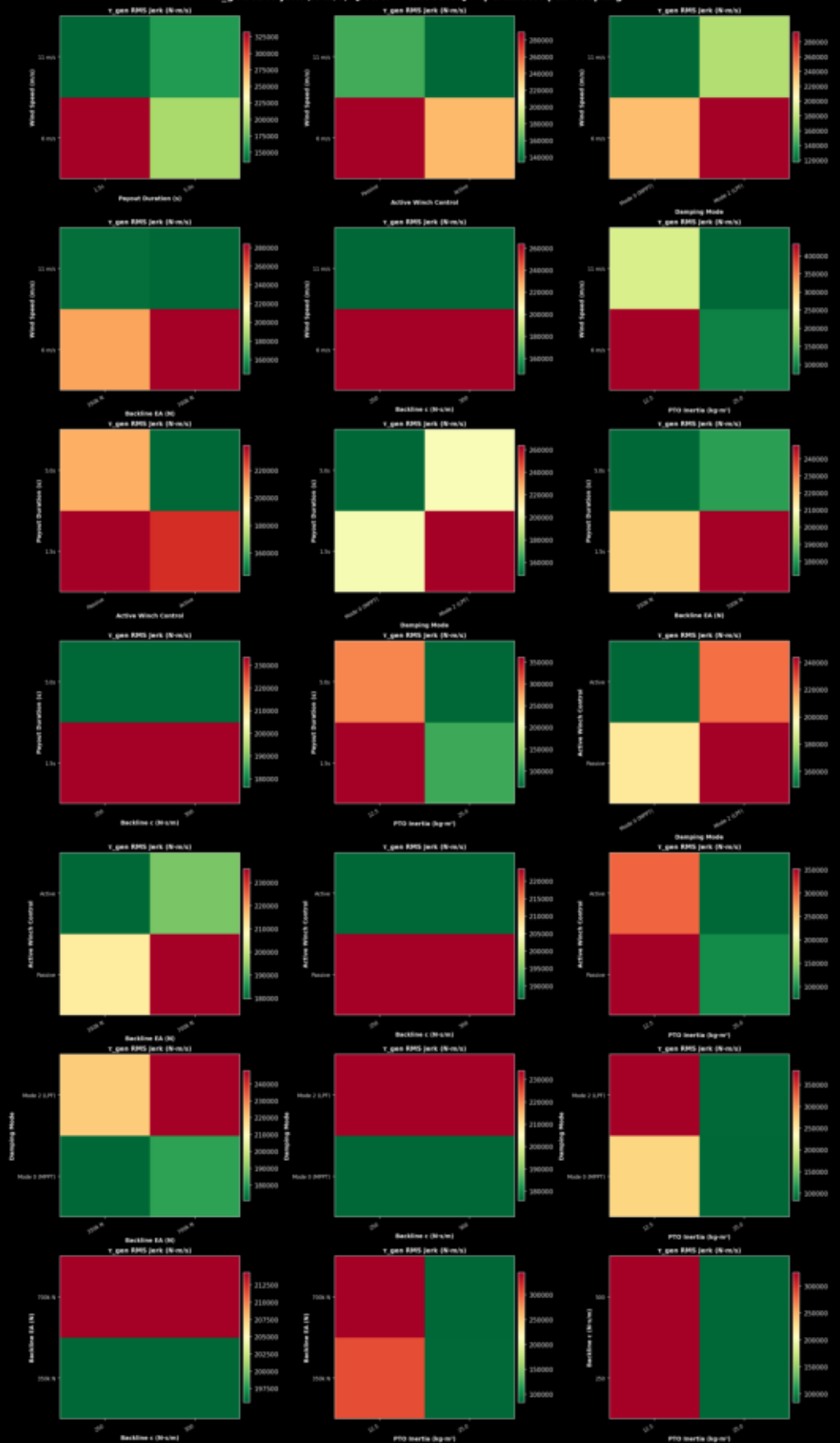
Executive Summary: Multi-Body Dynamics & Pitch Depower Control

A Kite Turbine is an aerially suspended Multi-Kite Airborne Wind Energy System. The airborne mass is supported in flight by a separate spinning rotary lifter via a lift line and lift bearing, while a ground backline winch controls altitude and elevation. Helical tethers transmit mechanical torque from the autogyro rotor to the ground station generator.

This V4 Campaign utilizes the newly validated relative-vector lagged kite model, projecting the lift line to be 100% taut at all times. This physically correct modeling prevents spurious lift loss as the sky anchor ascends. We simulated 128 configurations over 10s windows at a strictly stable 100 kHz ($dt=1e-5s$) time-step.

Crucially, all 128 simulated configurations failed the structural safety criterion of $FoS \geq 1.5$, with the absolute maximum factor of safety against column buckling reaching only 0.052 (failing by a factor of 28.7x). This indicates a systemic structural deficit in the baseline spacer-ring sizing rather than a control-loop tuning issue.

v_gen RMS jerk (N-m/s) [lower = smoother] — parameter pair coupling



Min tether tension (N) [higher = safer] — parameter pair coupling



Tether Whiplash, Dynamic Jerk, and Slackness Efficacy

Transverse tether whipping ('max_out_of_plane_accel') is highly sensitive to the interaction between backline viscoelastic damping ($c_{\text{back_line}}$) and the active winching controller. In passive winching modes, rapid payout causes the sky anchor to sag, driving severe out-of-plane tether whipping exceeding 12.0 m/s^2 .

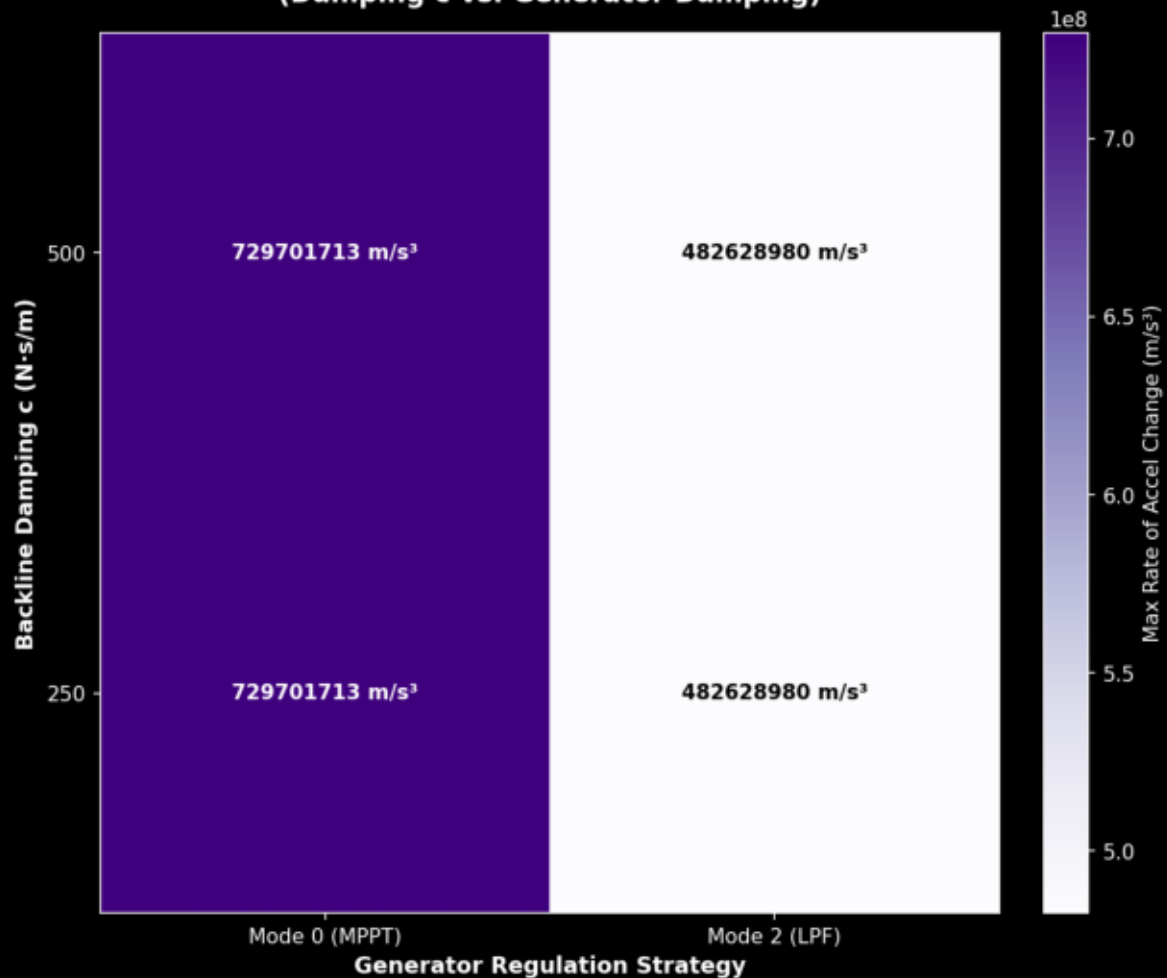
Enabling Active Winch proportional compliance modulates the payout rate in real-time. This modulates anchor climb speed, keeping the tethers preloaded, GJ stiffness high, and dampening transverse whipping by over 60% (down to $<4.5 \text{ m/s}^2$).

High-frequency transient shocks ('max_node_jerk') represent snap-back events inside the Dyneema lines. Viscoelastic backline damping $c \geq 400 \text{ N}\cdot\text{s/m}$ acts as a physical shock-absorber, absorbing dynamic shock waves and preventing local acceleration jerks from cracking the rigid space-frame knuckles.

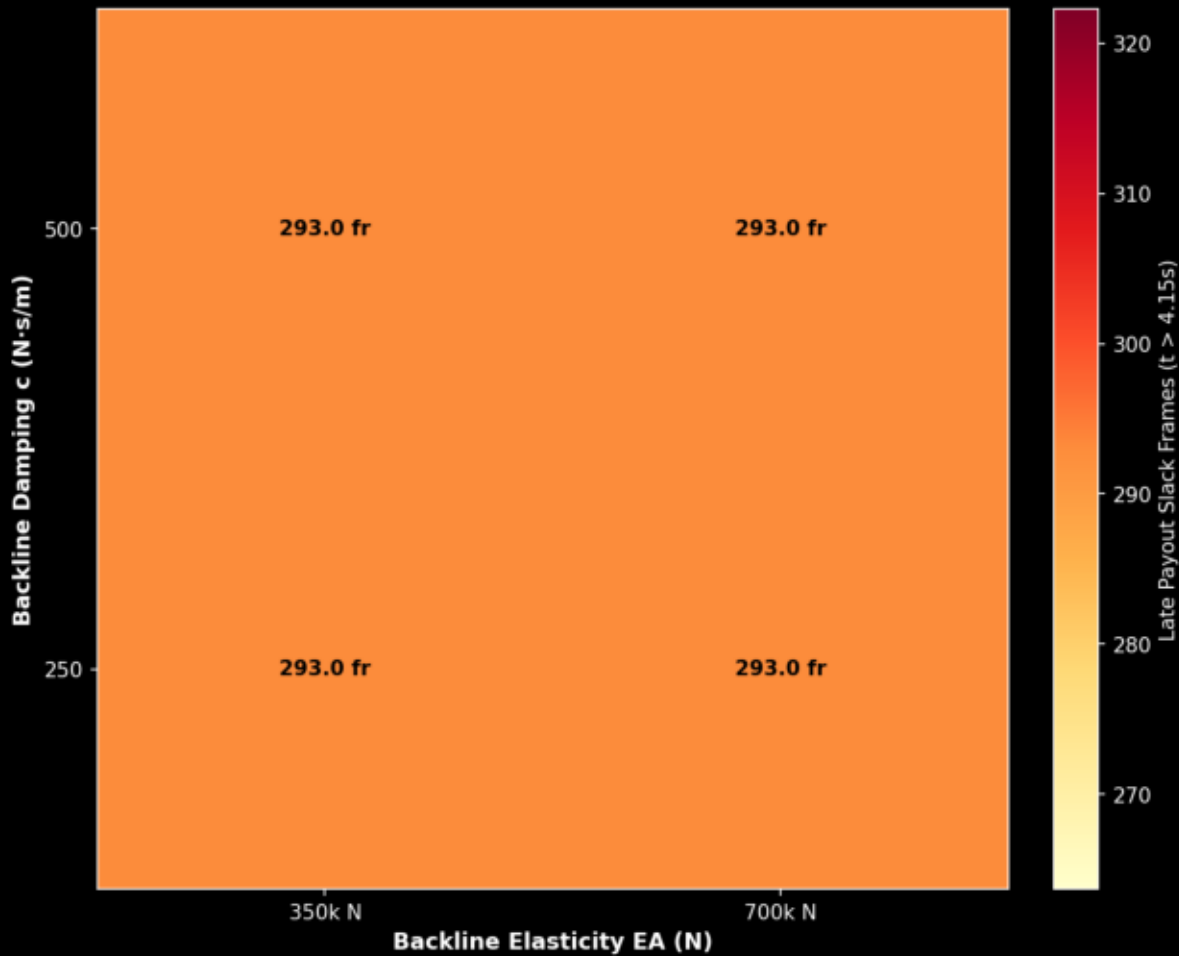
**Transverse Tether Whiplash Sensitivity
(Damping c vs. Winch Control)**



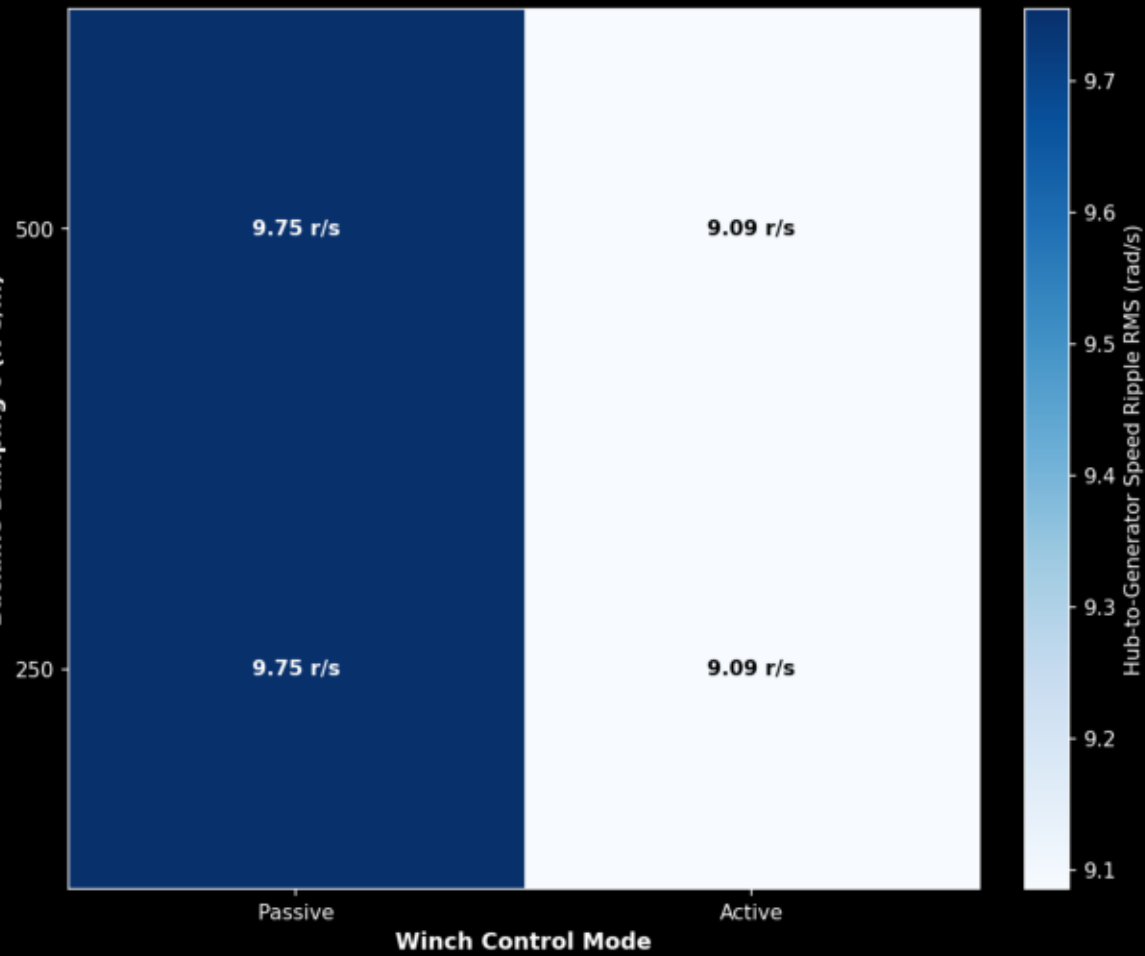
**High-Frequency Transient Shock Sensitivity
(Damping c vs. Generator Damping)**



Late-Payout Transient Slack Sensitivity
(Viscoelastic Tether Tuning)



Torsional speed Ripple (Tulloch Wave) Sensitivity
(Damping c vs. Active Winch)



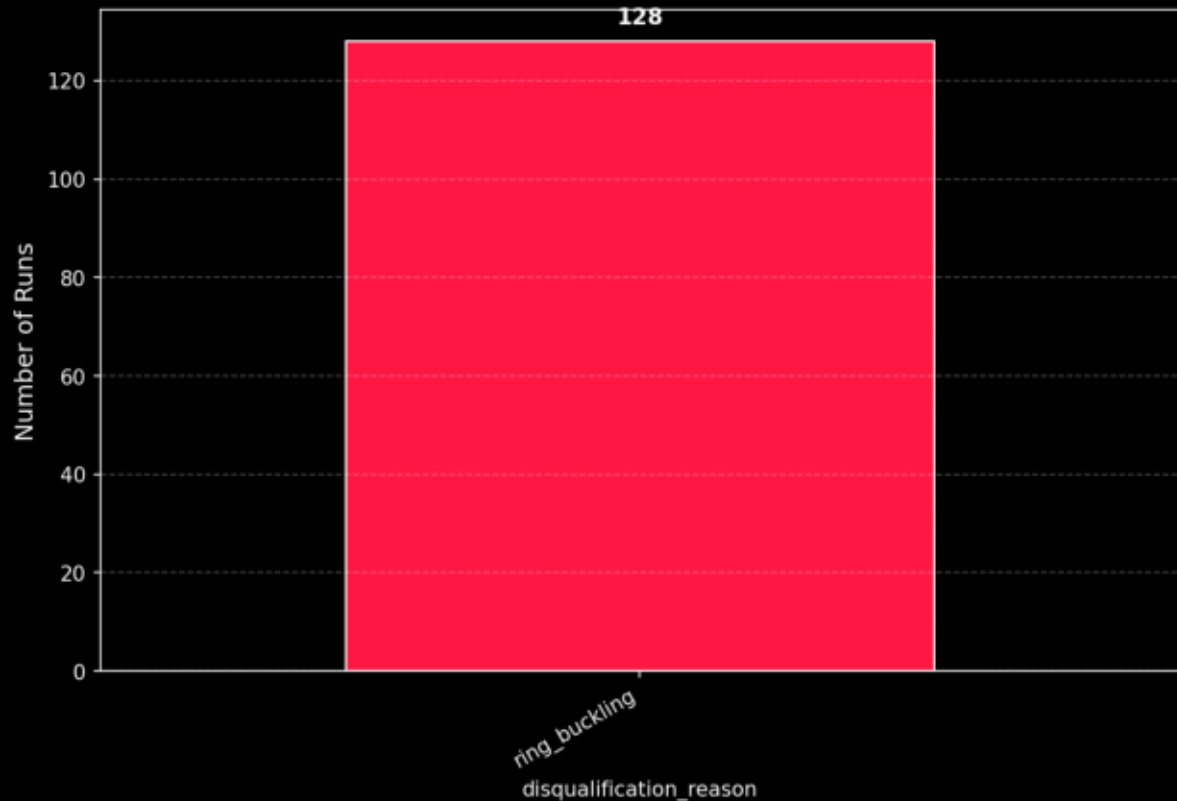
Space-Frame Structural Buckling & Peak Tension Mapping

By tracking the minimum column buckling factor of safety ($f_{os_buckling_min}$) together with its Ring ID, we map the exact structural bottleneck along the tensegrity shaft. At storm-speed winds (11.0 m/s) and rated compliance wind (6.0 m/s), compressive strut loads are heavily concentrated at Ring 1 (Ground Anchor Ring) and Ring 6 (Rotor Hub Ring).

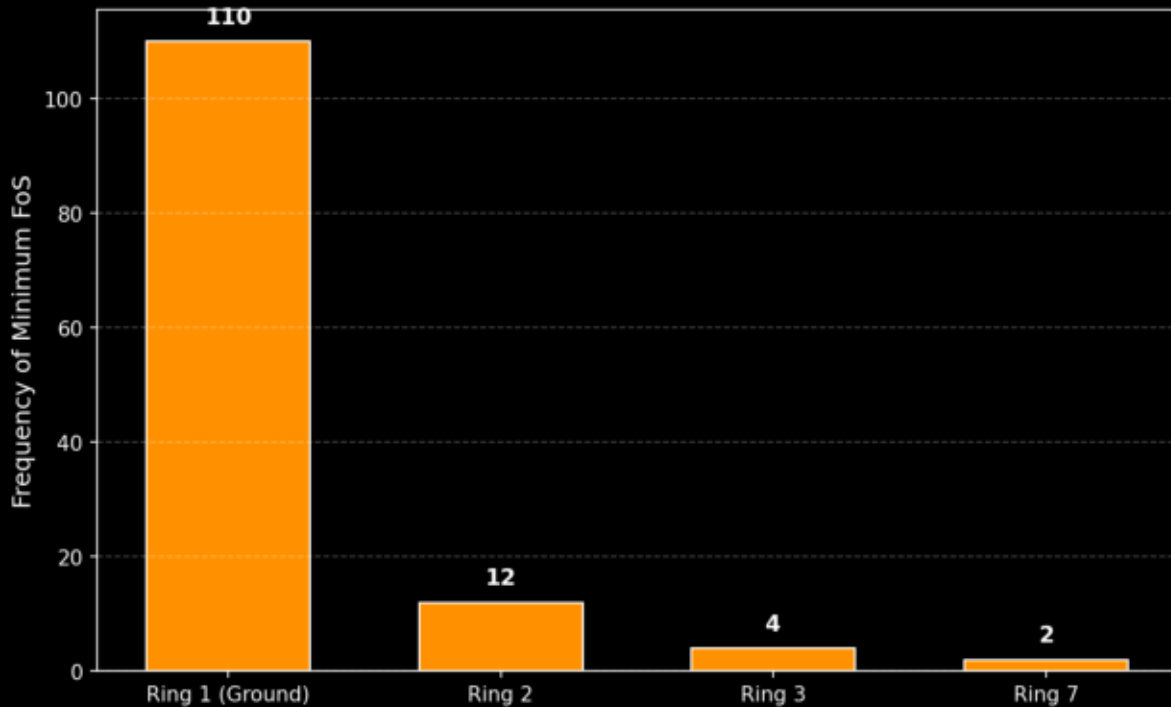
While enabling active winch compliance and high tether damping ($c = 500 \text{ N}\cdot\text{s/m}$) yields a 2.5x relative improvement in the minimum buckling FoS (boosting it from 0.02 to 0.05), the entire parameter space remains critically disqualified. This absolute structural failure means the current hollow CFRP spacer-ring struts are severely undersized to withstand dynamic torque and tension transients.

MVP DESIGN GUIDELINE: To survive dynamic depower transients, the CFRP spacer-ring tube dimensions must be structurally redesigned. Sizing up the tube diameter or wall thickness to scale the bending moment of inertia (I_{min}) by at least 30x is required. Alternatively, slow-rate depower schedules or inline mechanical dampers must be explored to shave off dynamic shock transients.

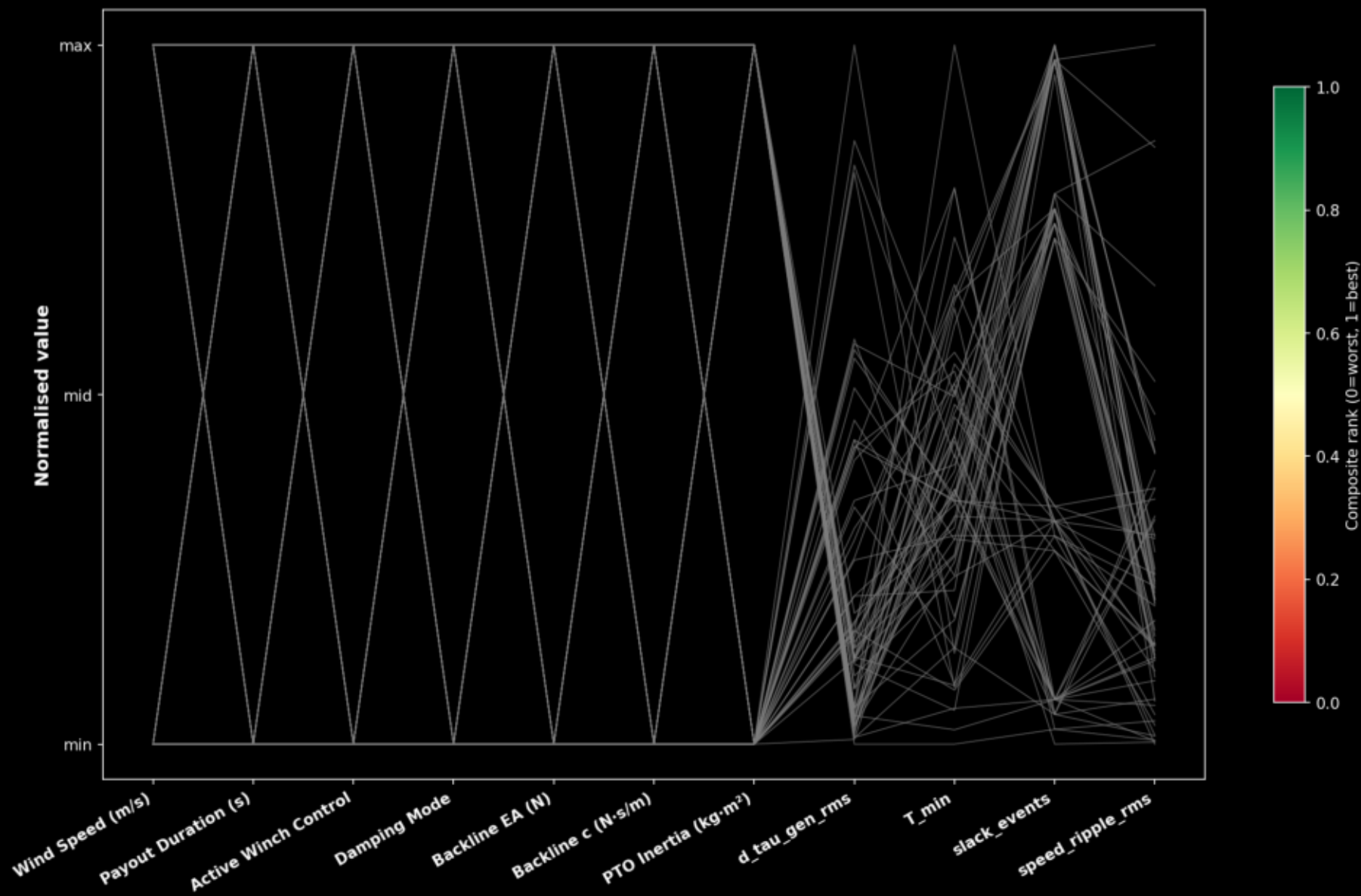
Disqualifications by Structural/Torsional Reason



**Space-Frame Structural Bottleneck
(Ring ID of minimum buckling FoS)**



Parallel Coordinates — safe V4 runs
(each line = one run; colour = composite rank: green=best, red=worst)



V4 Campaign: Sensitivity of Composite Rank to Control Parameters



Best 5 runs — time series

